Kalpana

An AI-Driven Mental Health Peer Support Platform

Empathy at 3 AM. No waitlists. No judgement.

Built with local AI + Sarvam Voice API

Mental Health Support Has a Critical Gap

1 in 8

People globally live with a mental health condition

3–6 Weeks

Average therapist waitlist — but distress doesn't wait

Scripted Bots

Existing chatbots miss emotional nuance

Power of Peers

Those who've survived are the most powerful support — but no matching system exists

An AI That Listens, Understands, and Connects

Listens

Real-time empathetic AI conversation. Speak or type — Kalpana understands both.

Understands

A silent clinical AI runs in the background, building a psychological profile from what the user shares.

Connects

When ready, it anonymously matches the user with a real human peer who has lived through the exact same experience — and lets them schedule a conversation.

Two AI Agents. One Conversation.

🎙 The Listener

ministral-3b, local via Ollama

- **Streams empathetic responses word-by-word in real time**
- **Dynamically shifts through 5 phases: Greeting → Explore → Probe → Process → Crisis**
- **Bases transitions on what the user shares**

🧠 The Clinical Mapper

gemma3:4b, local via Ollama

- **Runs silently in a background thread on every turn**
- **Outputs structured clinical profile: primary emotion, risk score (1–10), self-harm flag, root cause**
- **Both agents run concurrently via Python ThreadPoolExecutor — zero added latency**

Talk to Kalpana. Kalpana Talks Back.

01

User Speaks

Browser records audio (webm) via MediaRecorder. Sent to Sarvam's saaras:v3 Speech-to-Text API. Auto-detects language. Returns English transcript for the AI.

02

AI Responds

The Listener processes the transcript as normal and streams a text reply back.

03

Kalpana Speaks Back

The reply is translated to the user's detected language via Sarvam Translate, then synthesized to audio via Sarvam bulbul:v3 TTS (voice: Ritu). Audio plays inline in the chat.

Supports 9 languages: Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Bengali, and English.

What Powers Kalpana

Local / Backend

- FastAPI + Uvicorn
- Python
- LangChain
- Ollama (ministral-3b + gemma3:4b)
- HuggingFace SentenceTransformers
- Pinecone Serverless Vector DB
- Python ThreadPoolExecutor

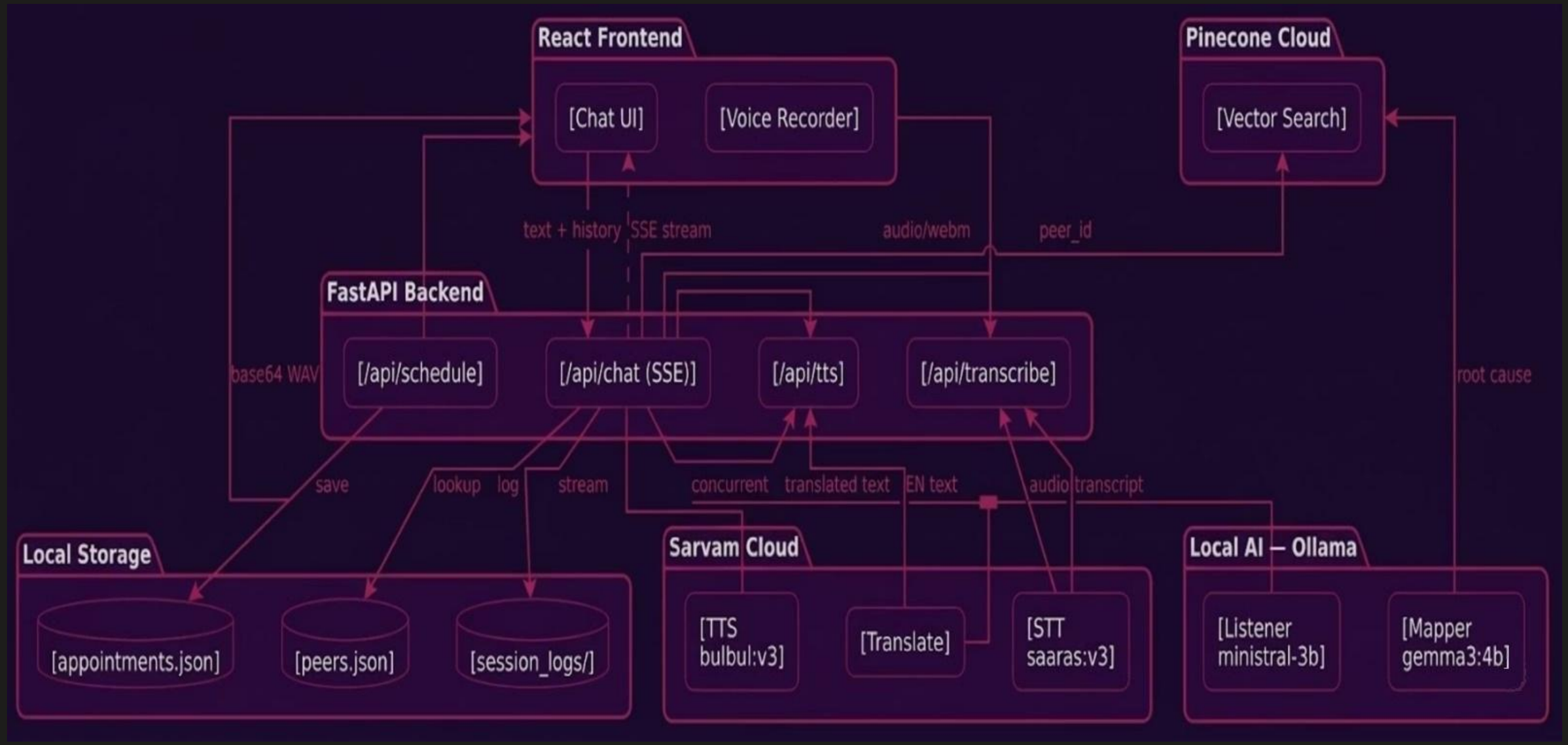
Cloud / Voice

- Sarvam AI
- STT: saaras:v3
- Translate: sarvam-translate:v1
- TTS: bulbul:v3

Frontend

- React 18
- Vite
- Tailwind CSS v4
- Server-Sent Events (SSE)
- MediaRecorder API

System Architecture



From AI Chat to a Real Human Connection

Threshold Met

After 2+ turns, risk score ≥ 5, and a root cause is identified

Vector Search

Root cause is embedded locally and matched against Pinecone (cosine similarity ≥ 0.70 required)

Match Found

A pop-up shows the peer's lived experience and available time slots

Booked

User picks a slot; appointment saved via POST /api/schedule

Safety Is Non-Negotiable

Immediate Crisis Intercept

If the Mapper detects self-harm indicators OR risk score ≥ 8 → crisis intercept triggers immediately

Direct Emergency Access

Peer scheduling is bypassed — no match is offered. A full-screen Crisis Modal replaces the UI with live helpline resources and emergency contacts

Session Safely Paused

Chat input is locked — the session is safely paused

Deterministic Code

This logic is deterministic Python code — never delegated to an LLM

What's Next

Real-Time Peer Chatrooms

WebSocket rooms that open directly after a peer slot is booked, enabling live text chat between user and peer

AI Chat Moderation

A lightweight background safety agent to passively monitor live peer sessions for harmful ideation

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> ***Kalpana is built for everyone who needed someone at 3 AM.***